

POKHARA UNIVERSITY

Level: Bachelor
Programme: BCA

Semester: Spring

Course: Probability and Statistics (New)

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) The weekly salaries of 84 workers in a factory are given below: 8

Salary	300-310	310-320	320-330	330-350	350-370	370-410
No. of workers	8	10	20	18	16	12

Construct a histogram, frequency polygon and frequency curve.

- b) One hundred and twenty students appeared for certain test and the following marks distribution was obtained. 7

Marks	0-20	20-40	40-60	60-80	80-100
No. of Students	10	30	36	30	14

Find:

- The limits of marks of middle 40% students.
- The percentage of students getting marks more than 75.
- The number of students who fail, if 35 marks are required for passing.

2. a) The following are the numbers of minutes that a person had to wait for the bus to work on 15 working days, 10, 1, 13, 9, 5, 9, 2, 10, 3, 8, 6, 17, 2, 10, 15. Draw a box plot and interpret the result. 8

- b) A factory has 3 units A, B and C. A produces 25% of its products, B produces 25% and C produces 50%. The percentage of defective items produced by these units A, B and C are 1%, 2% and 3% respectively. If one of them is selected randomly from the total and found to be defective, what is the probability that it is produced by the machine A, B and C? 7

OR

State and prove Baye's theorem.

3. a) A problem in statistics is given to three students A, B and C whose chances of solving it are $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{1}{5}$ respectively. Find the probability that 5

- The problem will be solved.
- Only one of them can solve the problem.
- None of them will solve the problem.

- b) In a binomial distribution with 6 independent trials the probabilities of 3 and 4 success are found to be 0.2457 and 0.0819. Find parameter p and q. 5
- c) If a sample mean is 20, standard deviation is 3 and sample size is 64. Find the interval estimate of the population mean. ($\alpha=5\%$). 5
4. a) A set of examination marks is approximately normally distributed with a mean of 75 and standard deviation of 5. If the top 5% of students get grade A and the 25% get grade F, what mark is the lowest A and what mark is highest F? 7
- b) A random sample of 100 items taken from a large batch of articles contain 5 defective items. Set up 95% and 99% confidence limits for the proportion of defective items in the batch. 8

OR

Explain the difference between the probability sampling and non – probability sampling.

5. a) A manufacture claims that the mean life of batteries manufactured by his company is 44 months. A random sample of 40 of these batteries was tested, resulting in a sample mean life of 41 months with a sample standard deviation of 9 months. Test at $\alpha=0.05$. 5
- b) The following are the average weekly losses of worker-hours due to accidents in 10 industrial plants before and after a certain safety program was put into operation: 5

Before	45	73	46	124	33	57	83	34	26	17
After	36	60	44	119	35	51	77	29	24	11

Use the 0.05 level of significance to test whether the safety program is effective?

- c) In an investigation into eye color and left-or-right-handedness the following results were obtained. 5

Eye color	Handedness	
	Left	Right
Blue	15	85
Brown	20	80

Is there evidence, at the 5% level of significance, an association between eye color and left-or-right-handedness.

6. a) Calculate Spearman's correlation coefficient from the following data: 7

X	48	43	40	9	16	16	65	24	16	57
Y	13	13	24	6	15	4	20	9	6	19

- b) The following data gives the experience of machine operators in years and their performance as given by number of good parts turned out per pieces. 8

Operator	A	B	C	D	E	F	G	H
Experience (X)	16	12	18	4	3	10	5	12
Performance (Y)	87	88	89	68	78	80	75	83

Calculate the regression equation of performance ratings on experience and estimate the probable performance if an operator has 8 years experience. Compute the coefficient of determination and interpret the result.

7. Write short notes on: (**Any two**) 2×5
- a) Difference between Primary and Secondary data
 - b) Random Variable
 - c) Criteria of Good Estimator